



Fundamentals of Machining

Groover M.P.: Chapter 17



The student knows how:

- understand the mechanism of chip formation;
- knows industrial machining processes;
- optimize costs in machining;
- improve process planning.





Goal: to remove material to transform the raw part into the desired geometry.

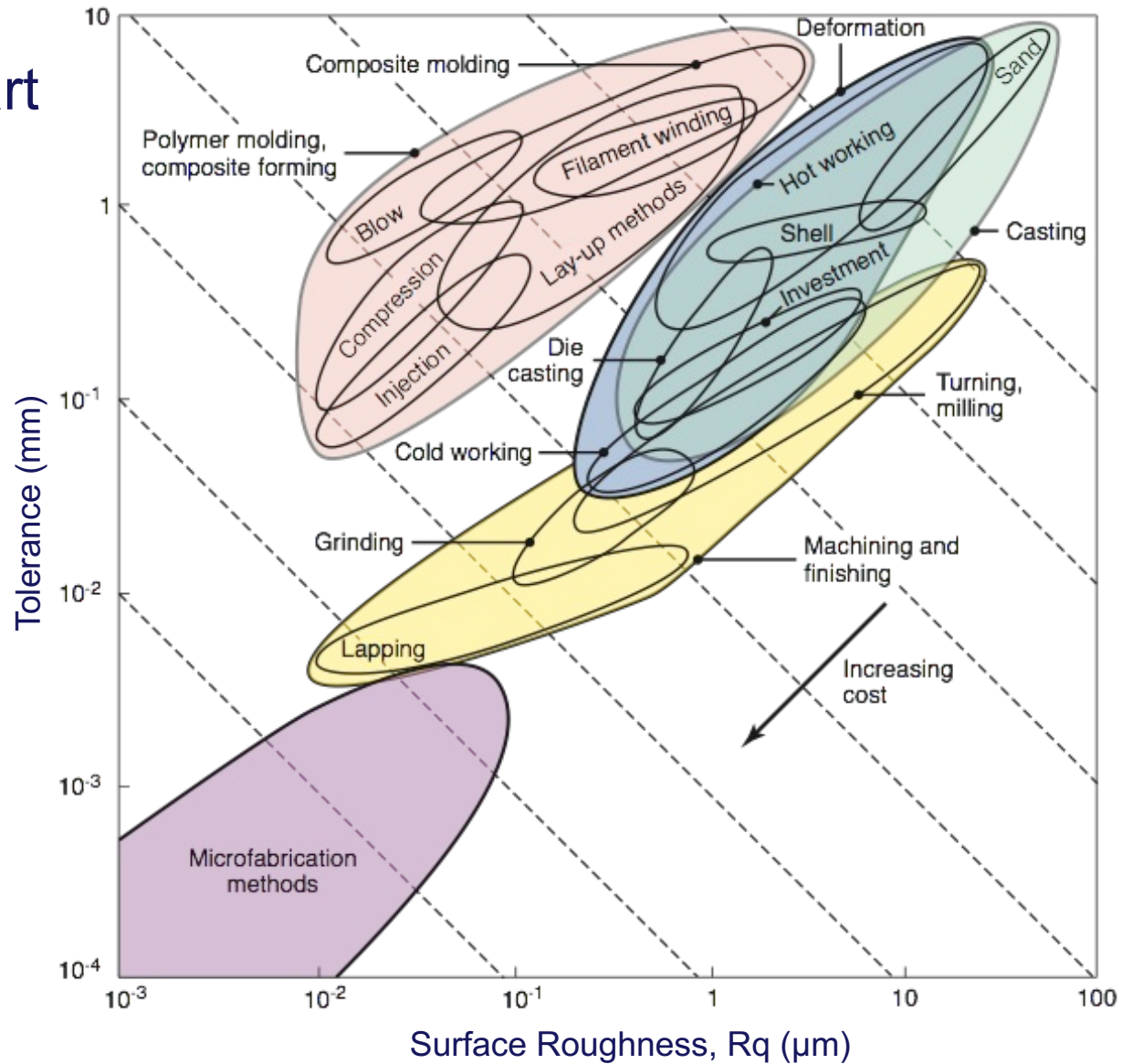
In general they are the last processes performed on the mechanical components. In fact, machining processes allow to obtain good tolerances and good surface finish.

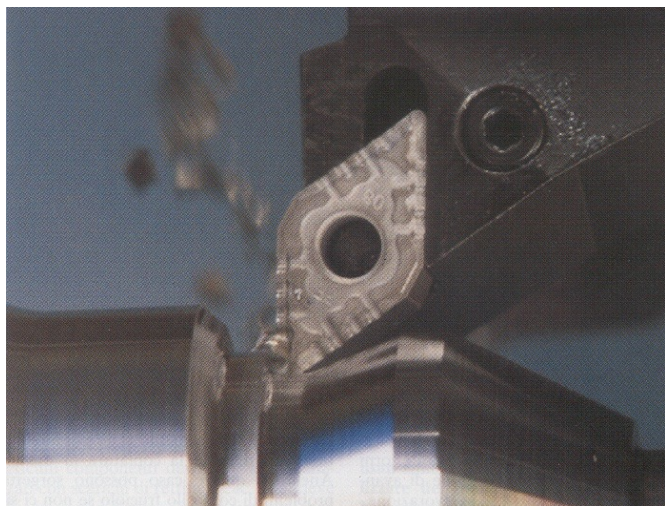


A forged crankshaft before and after machining the bearing surfaces. The shiny bearing surfaces of the part on the right cannot be made to their final dimensions and surface finish by casting or metal forming processes.



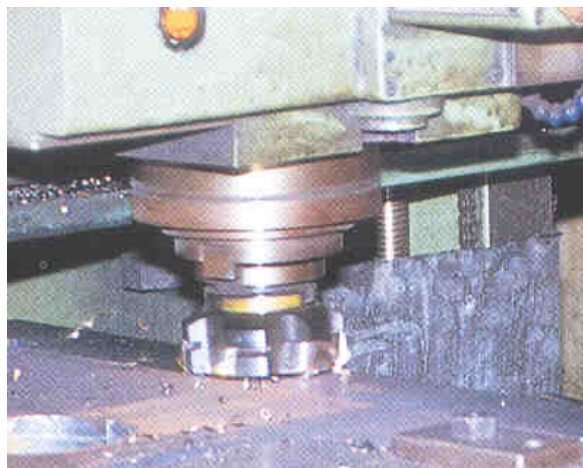
Ashby Chart





Turning

Drilling



Milling



We will study:

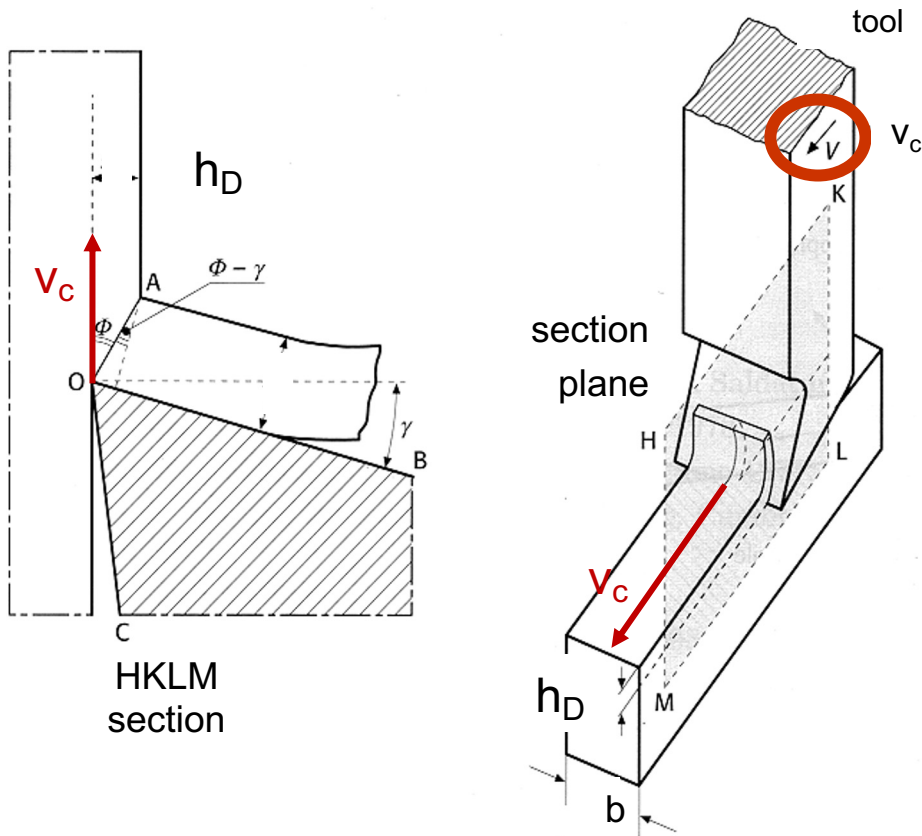
- ✓ **“Orthogonal” cutting**
a “simplified” industrial process

- ✓ **More complex industrial processes:**
 - Turning
 - Drilling
 - Milling
 - etc.



We study this machining process because:

- It is **simple** to describe from the kinematic (motions) and dynamic (forces) points of view;
- It allows us to **understand** the elementary mechanism of chip formation;
- Many of the **variables** met in orthogonal cutting are present in the more complex industrial processes (turning, milling, etc).



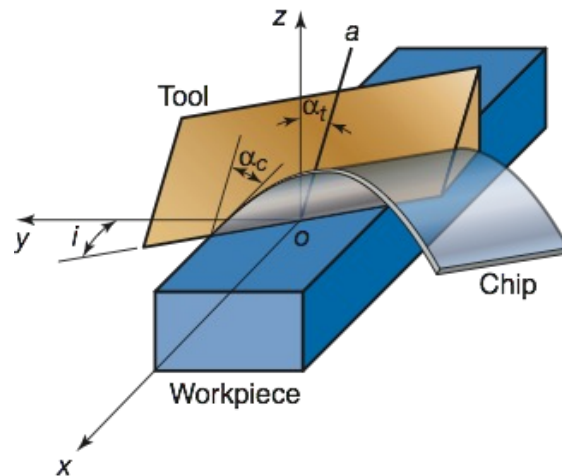
We have orthogonal cutting when the tool moves with a **cutting speed orthogonal to the cutting edge** and only the central portion of the cutting edge is involved in the cutting.

Therefore, all the kinematic and dynamic variables belong to a plane that is orthogonal to the cutting edge and to which the cutting speed belongs.

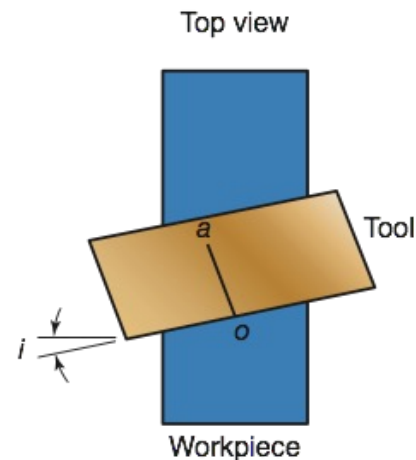


CUTTING WITH AN OBLIQUE TOOL

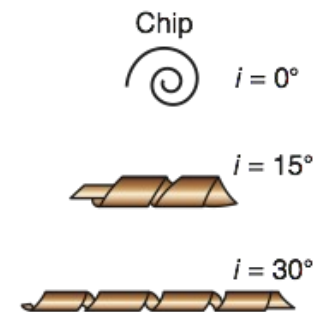
In practice, most of the cutting processes use oblique tools and the study of chip formation is quite complex.



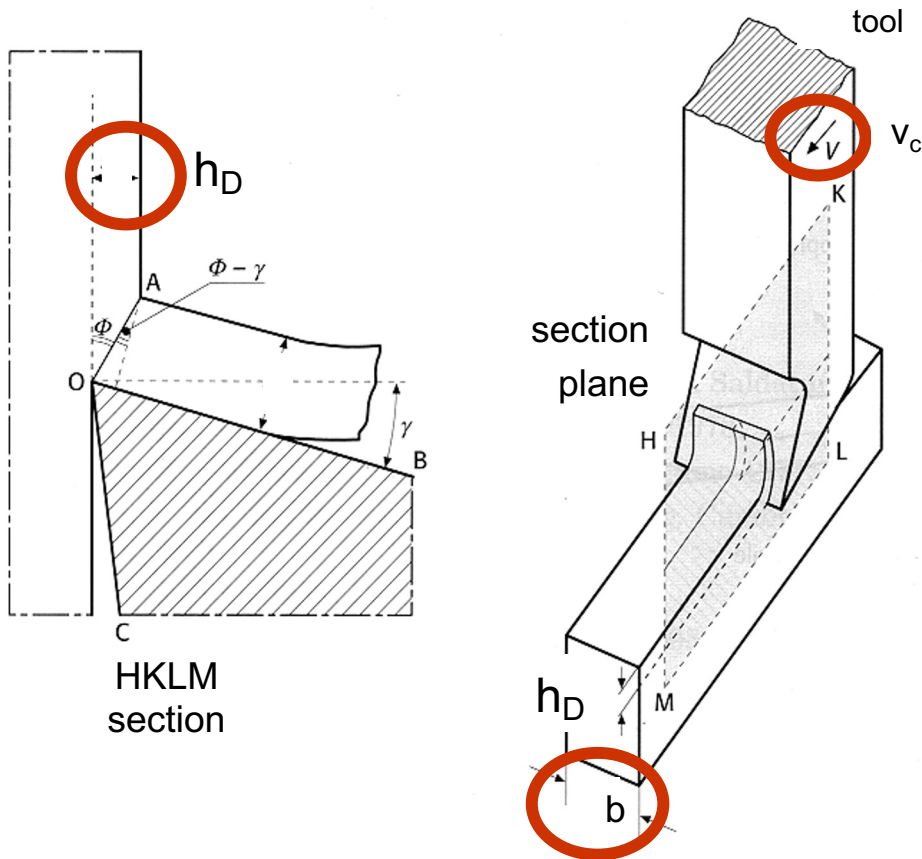
(a)



(b)



(c)



Relevant Variables:

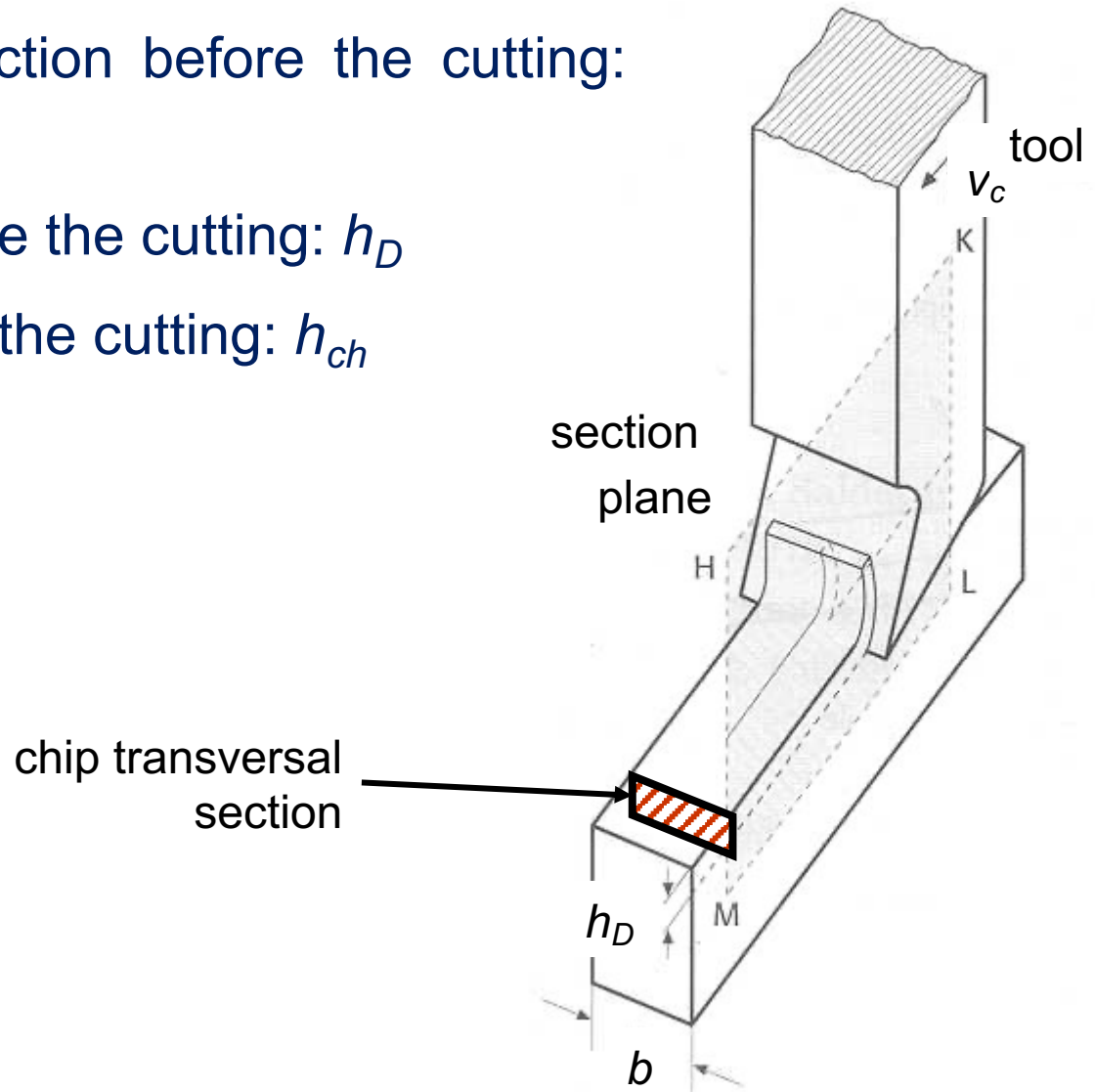
- v_c = cutting speed
- b = cutting width
- h_D = cutting thickness
(chip thickness before chip separation)



ORTHOGONAL CUTTING

- Chip transversal section before the cutting: $A_D = h_D b$
- Chip thickness before the cutting: h_D
- Chip thickness after the cutting: h_{ch}
- Chip thickness ratio:

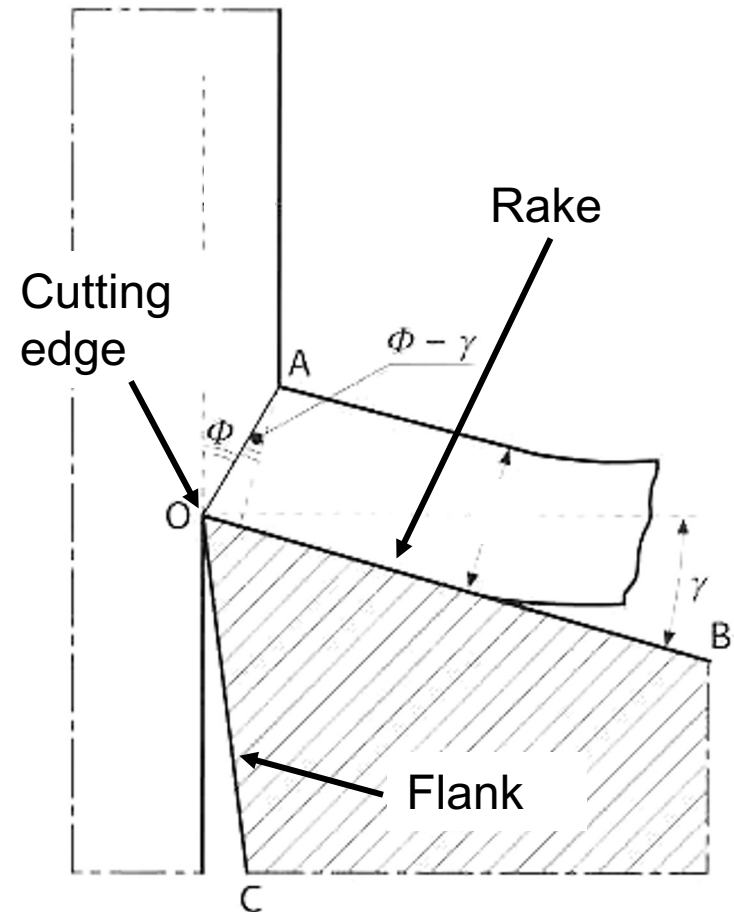
$$r = \frac{h_D}{h_{ch}} < 1$$





The tool is composed of:

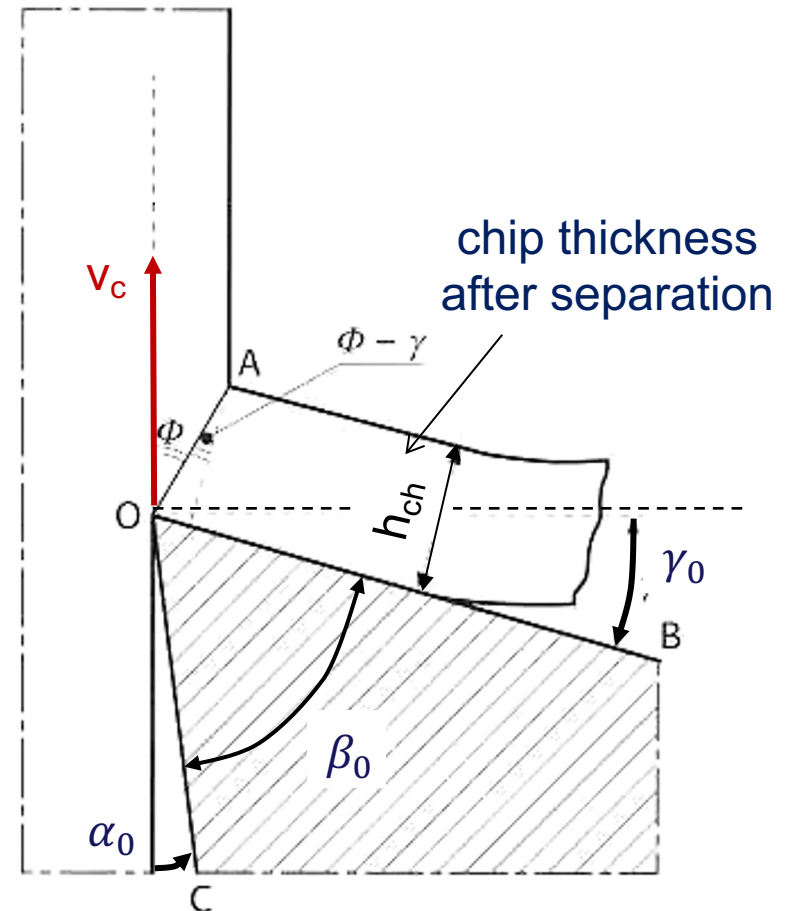
- **Rake face:** surface on which the chip flows.
- **Flank face:** surface looking at the machined surface.
- **Cutting edge:** intersection line between rake face and flank.





Cutting angles:

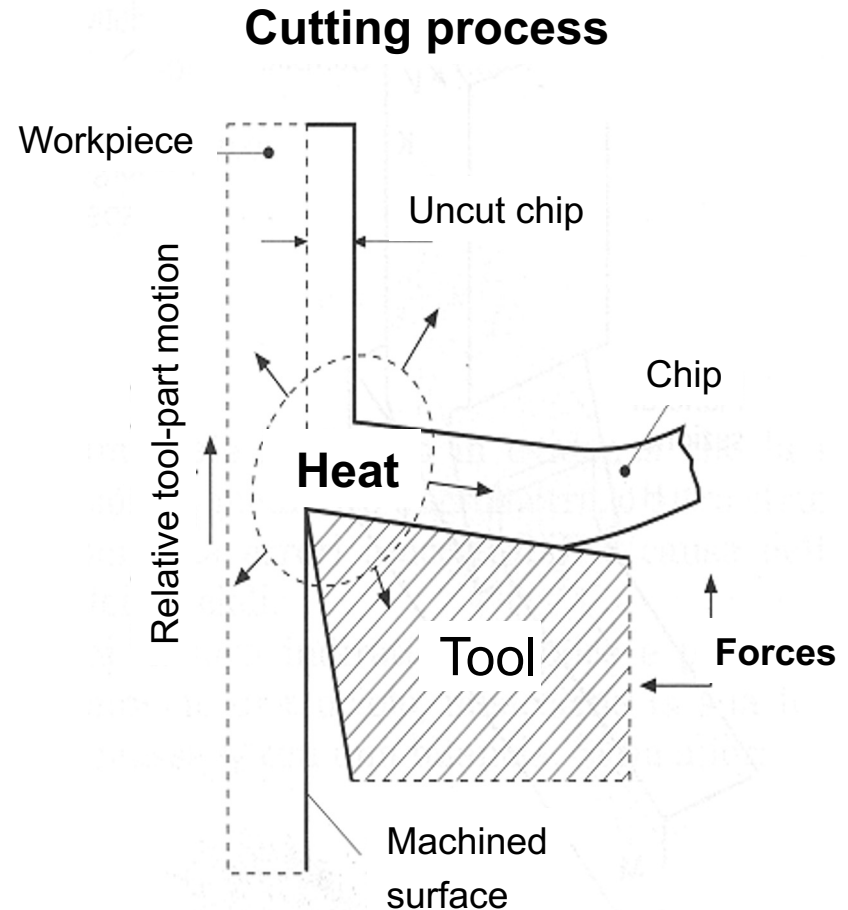
- **Rake angle γ_0**
Between the rake face and the normal to the cutting direction:
 $-15^\circ \leq \gamma_0 \leq 30^\circ$
- **Clearance (or relief) angle α_0**
Between the flank and the cutting direction:
 $2^\circ \leq \alpha_0 \leq 15^\circ$
- **Solid angle β_0**
Between rake and flank faces:
 $\alpha_0 + \beta_0 + \gamma_0 = 90^\circ$





MECHANISM OF CHIP FORMATION

- A relative motion between tool and workpiece exists.
- The tool stresses the material.
- The material is plastically deformed to fracture, separates from the workpiece and is transformed into chip.
- The chip flows on the tool and moves away from the machining zone.



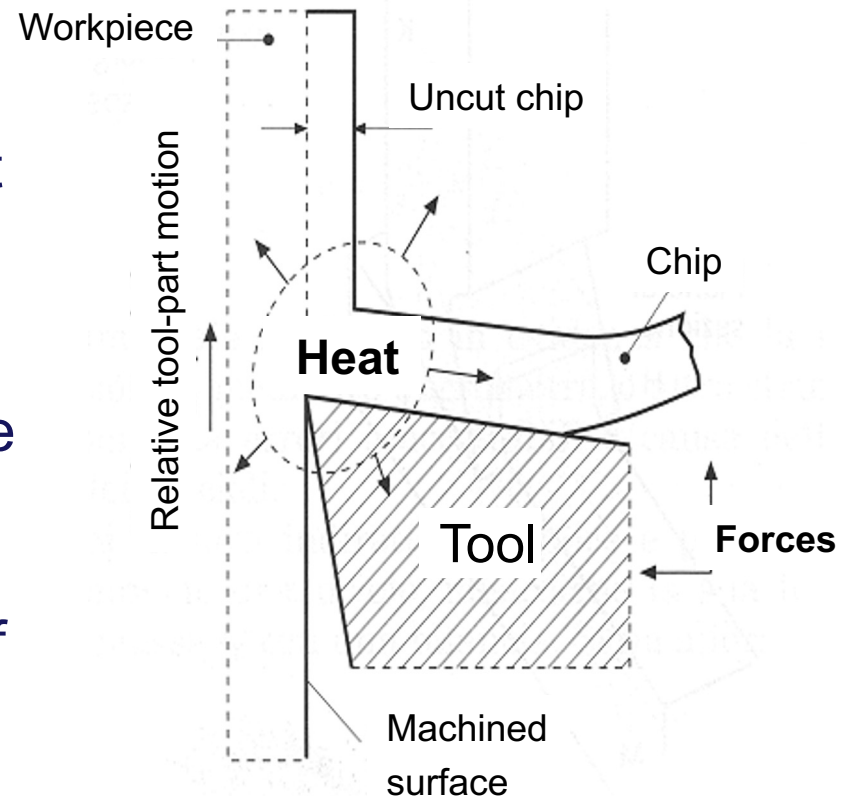


MECHANISM OF CHIP FORMATION

Associated to chip removal:

- **Forces:** Tool and workpiece exchange forces needed to deform the material, separate it from the workpiece and transform it into chip.
- **Heat:** *Plastic deformation* of the material before chip separation and *friction* between chip and tool generate a large amount of heat.

Cutting process



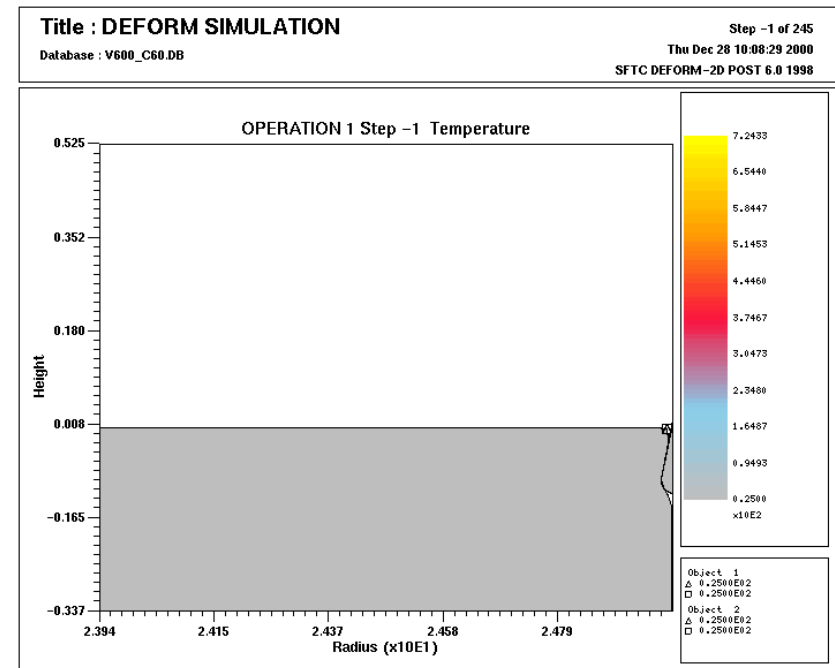
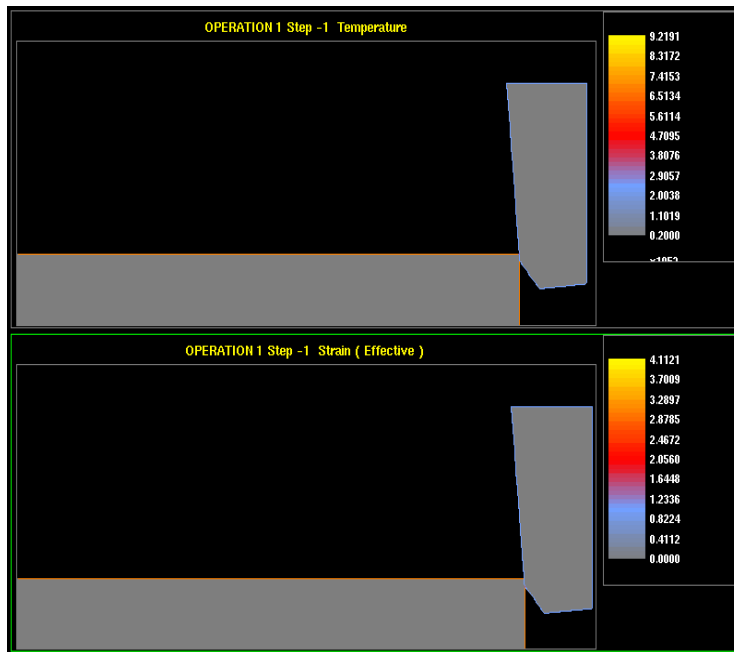


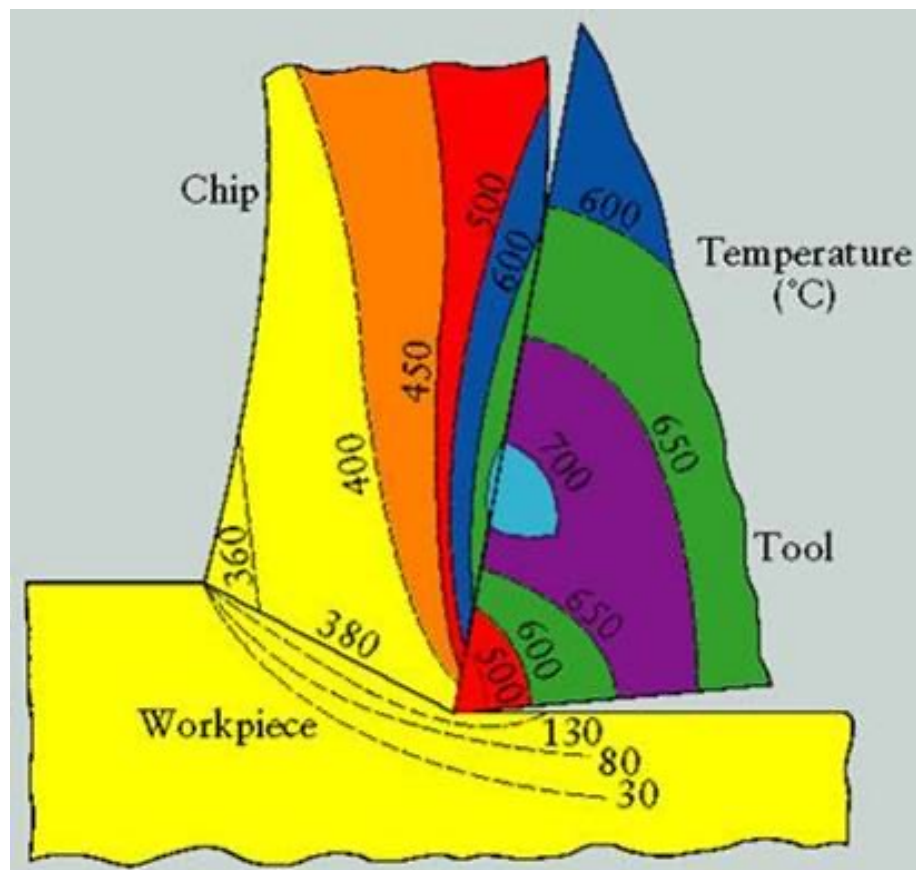
STRESS AND TEMPERATURE

16

temperature

strain





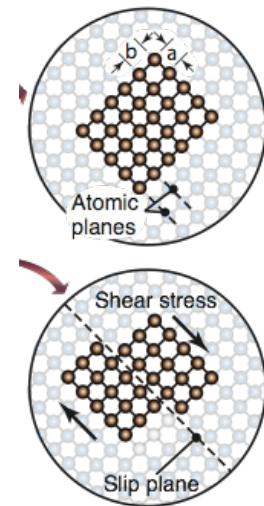


MECHANISM OF CHIP FORMATION

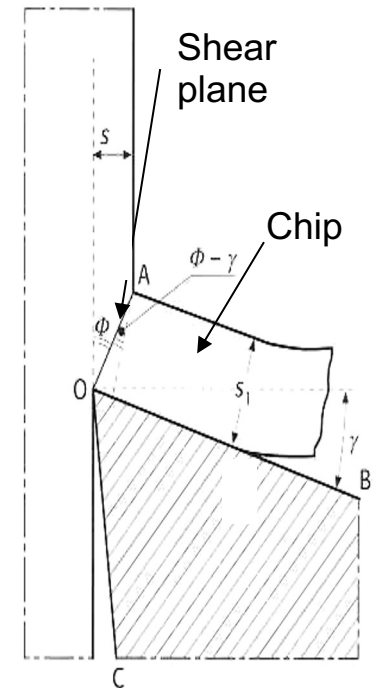
How does the material (*machining allowance*) transform into chip?

The tool stresses the material until this deforms plastically. With good approximation it can be said that the region where the material is deformed is a plane, called ***shear plane***.

The deformation proceeds until the separation of the chip from workpiece. Then, the chip flows on the tool.



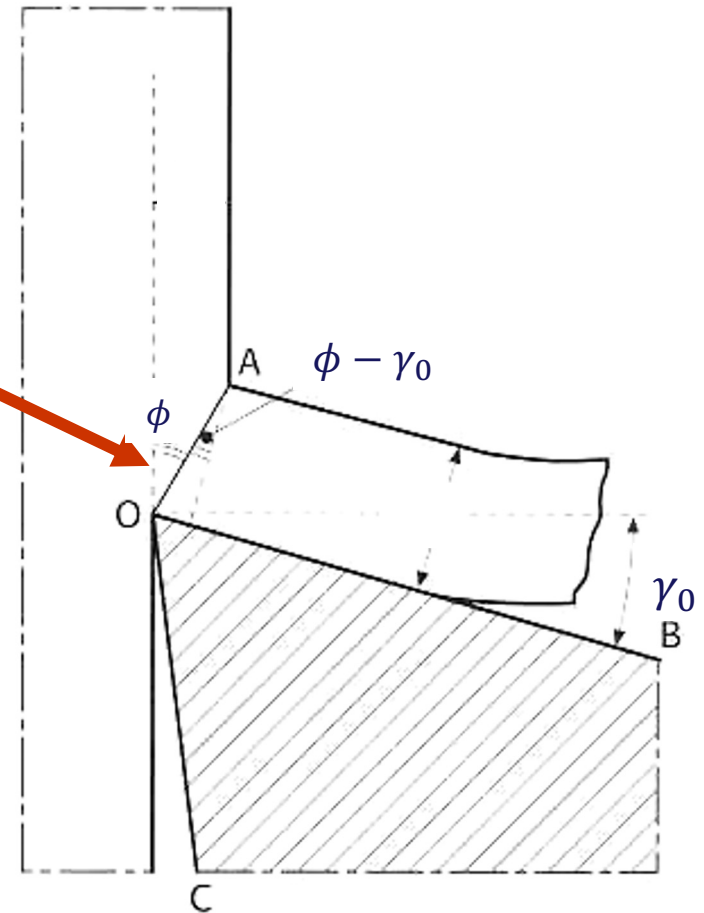
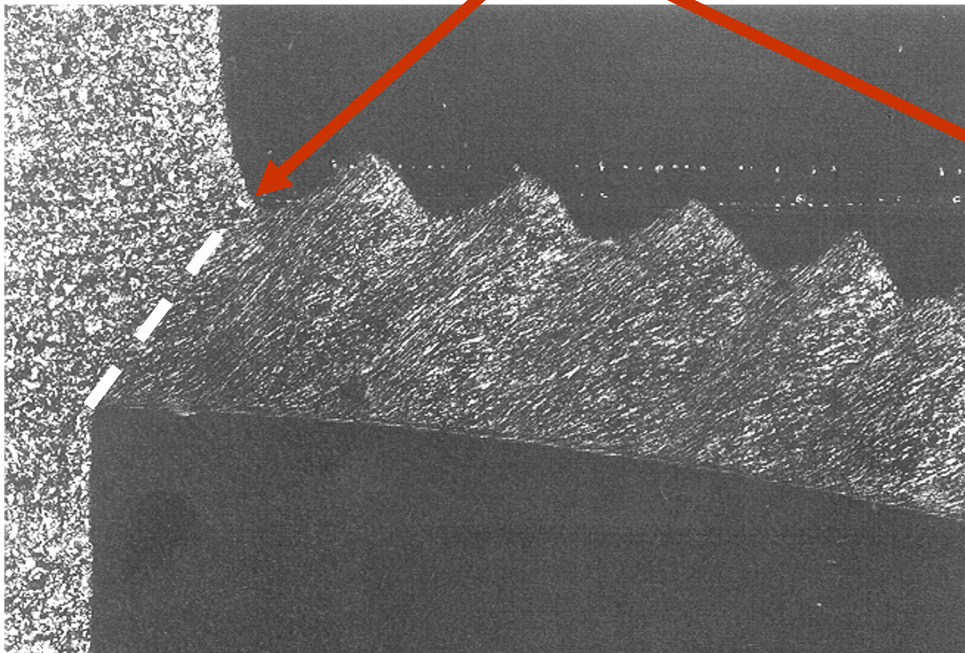
Shear stress





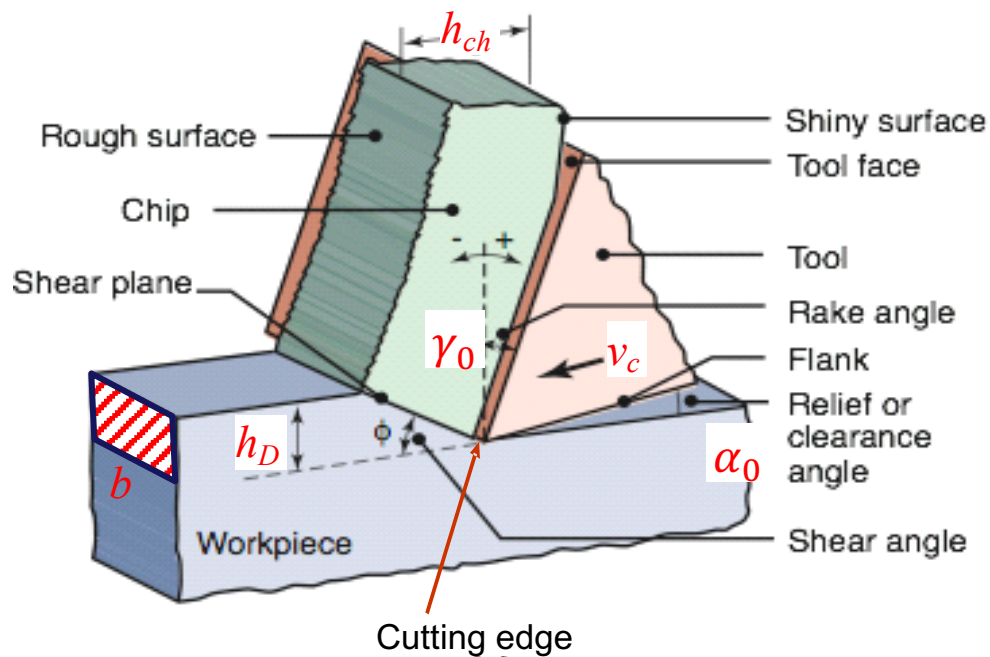
MECHANISM OF CHIP FORMATION

Shear plane

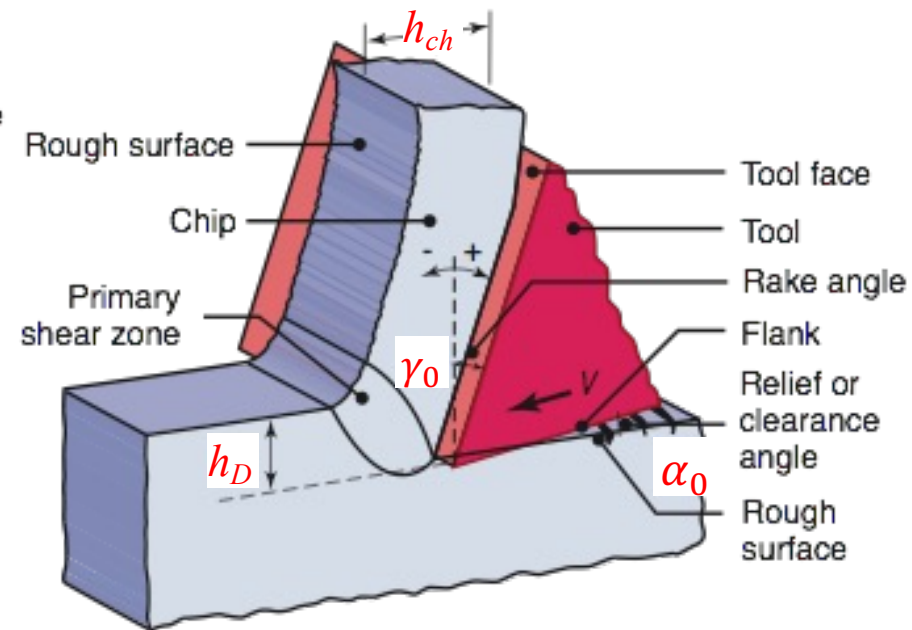




MECHANISM OF CHIP FORMATION



Shear plane model

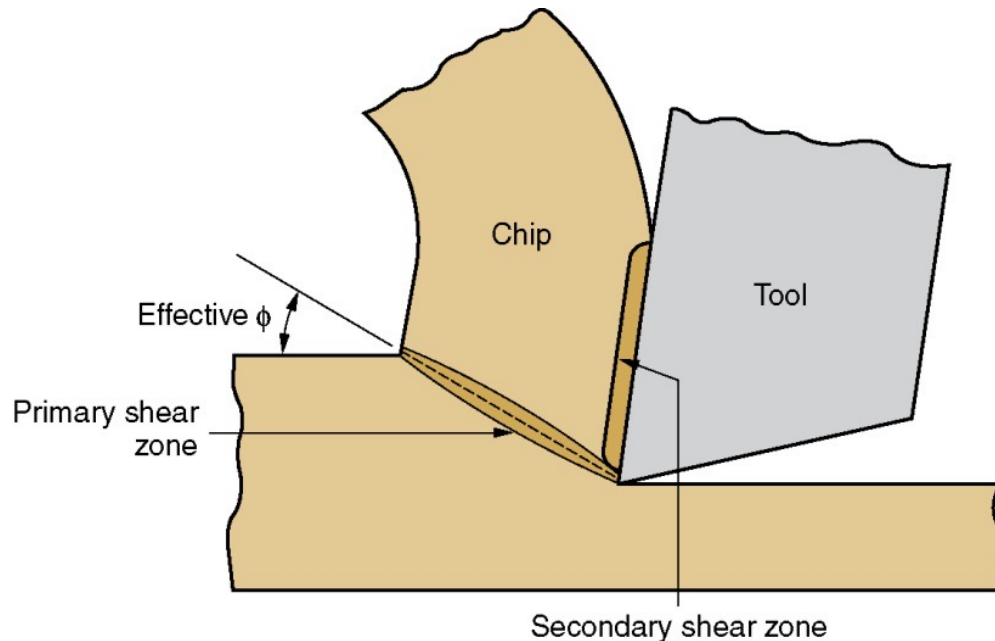


Shear zone model



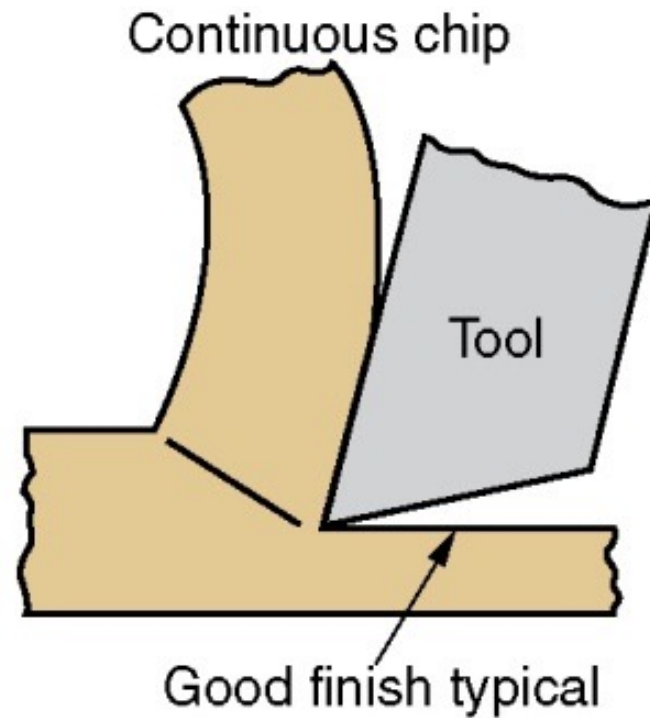
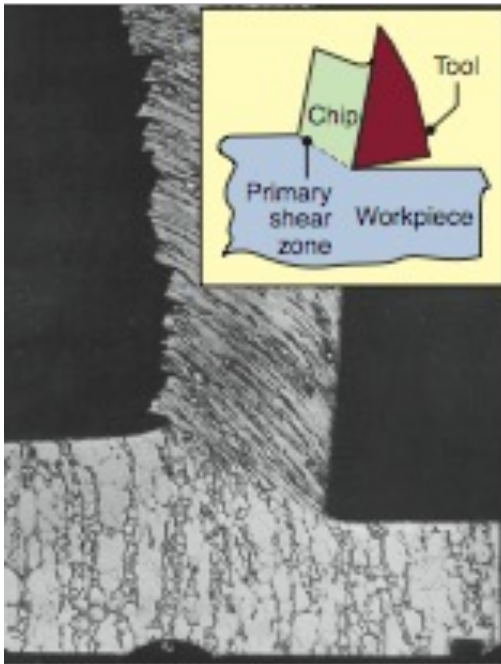
MECHANISM OF CHIP FORMATION

More realistic view of chip formation, showing shear zone rather than shear plane. Also shown is the secondary shear zone resulting from tool-chip friction.





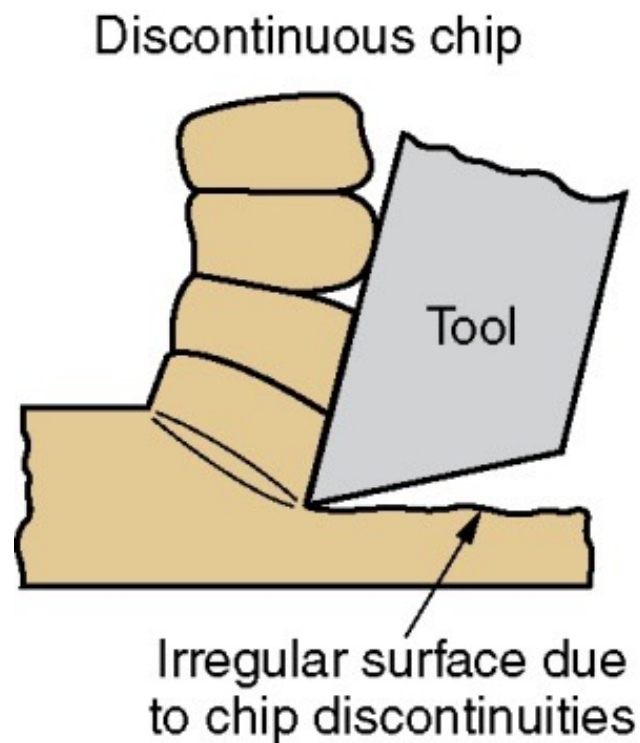
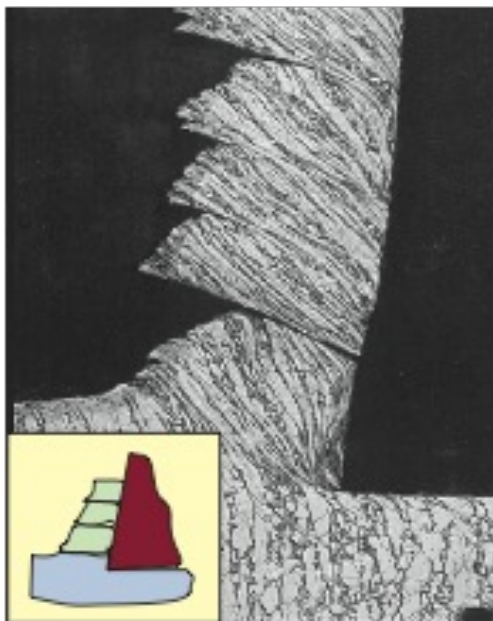
Continuous chip





MECHANISM OF CHIP FORMATION

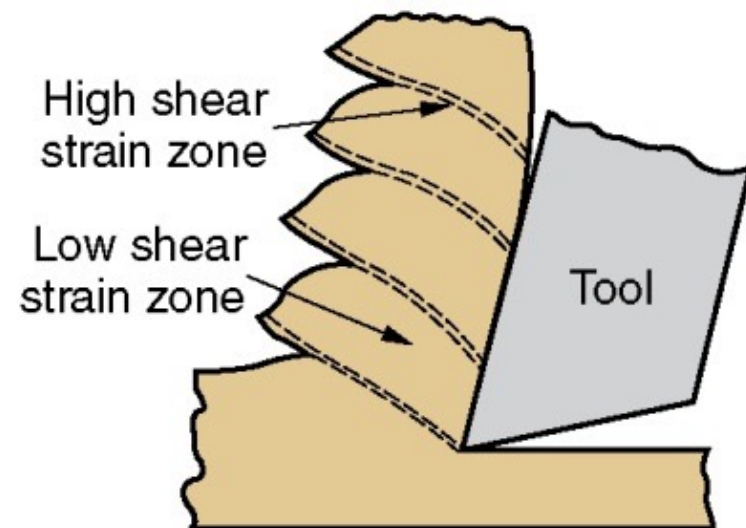
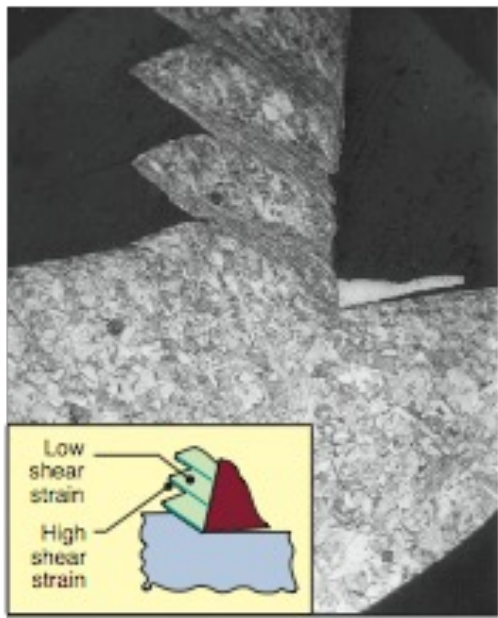
Discontinuous chip (brittle materials)





MECHANISM OF CHIP FORMATION

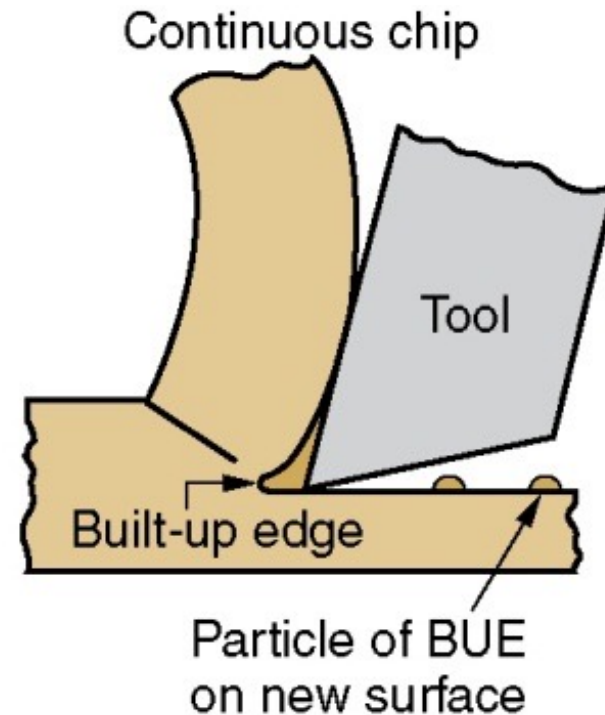
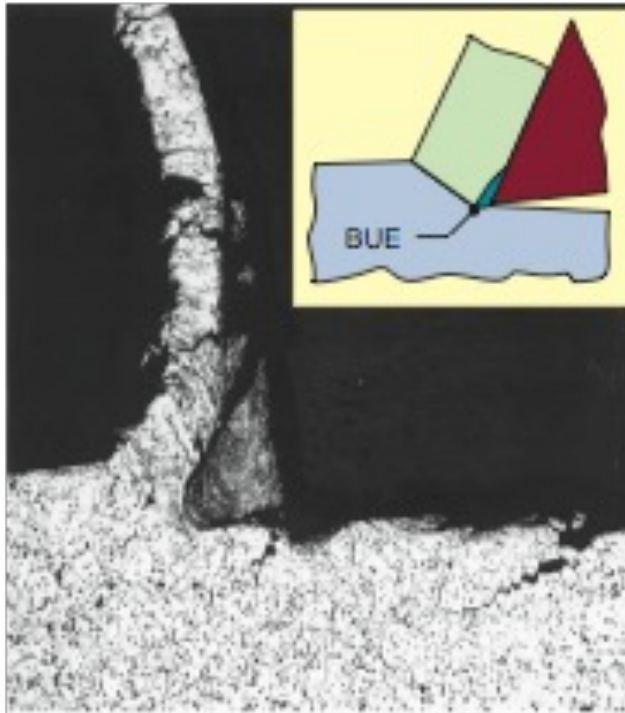
Serrated or segmented chip:
semi-continuous chip with saw-tooth appearance





MECHANISM OF CHIP FORMATION

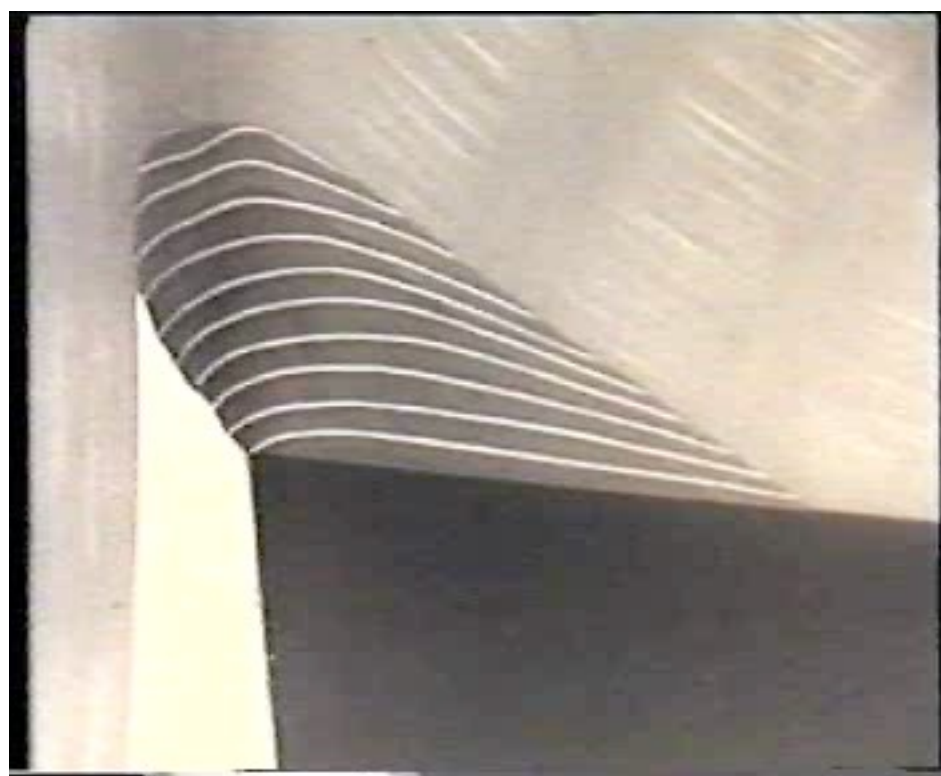
Continuous chip with built-up edge (BUE)





MECHANISM OF CHIP FORMATION

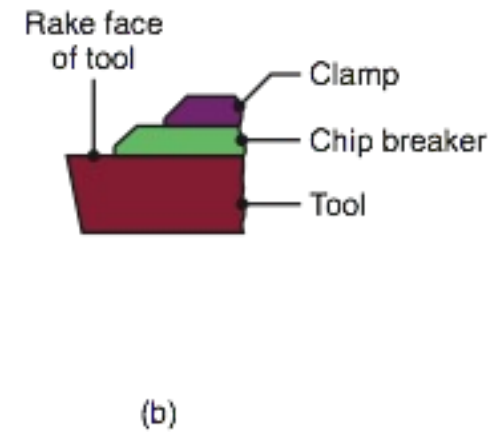
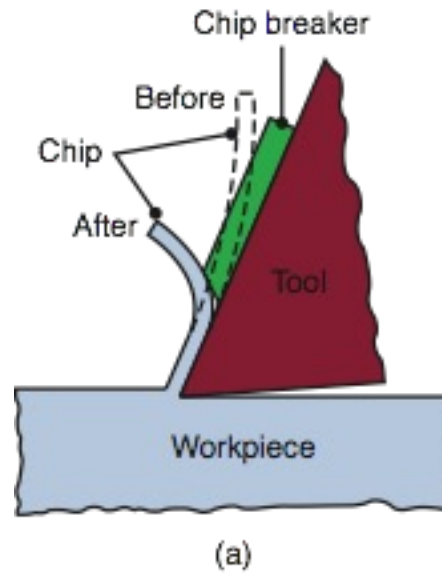
Continuous chip with built-up edge (BUE)



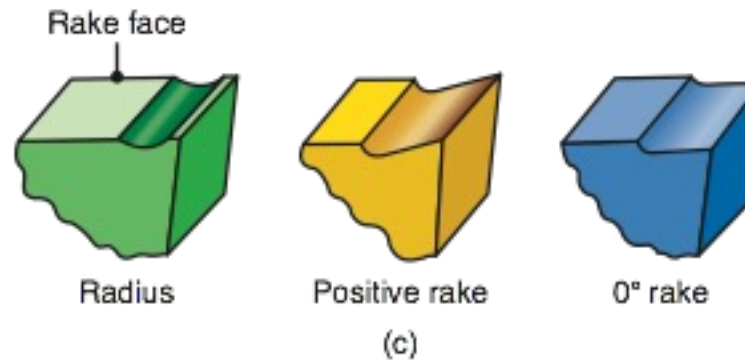


MECHANISM OF CHIP FORMATION

Chip breaker



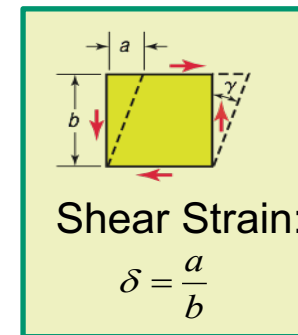
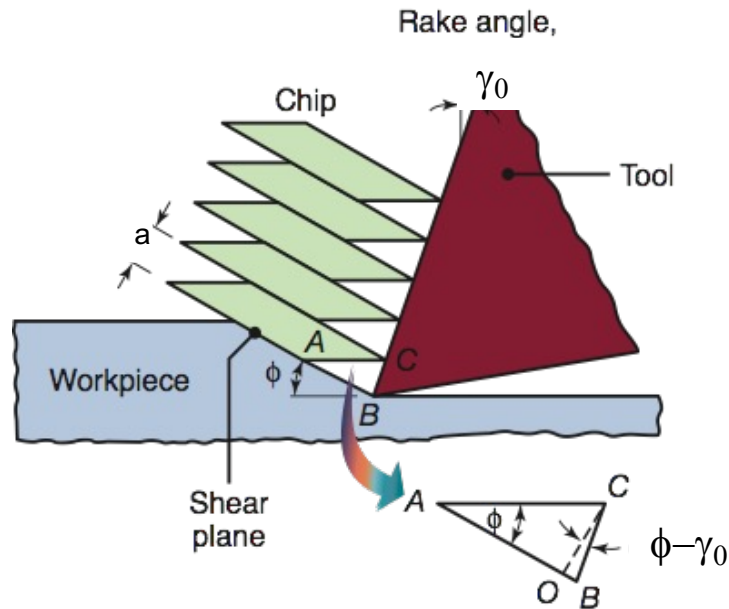
Grooves as
chip breaker





MODEL OF CHIP FORMATION

Card deck model of chip formation (Pijspanen, 1937)



Cutting ratio $r_c = \frac{h_D}{h_{ch}} = \frac{\sin \phi}{\cos(\phi - \gamma_0)}$

Shear strain $\delta = \frac{\overline{AB}}{\overline{OC}} = \frac{\overline{AO}}{\overline{OC}} + \frac{\overline{OB}}{\overline{OC}} = \cot \phi + \tan(\phi - \gamma_0)$



Cutting ratio

$$r_c = \frac{h_D}{h_{ch}}$$

$$r_c = \frac{h_D}{h_{ch}} = \frac{b_{ch} \cdot l_{ch}}{b_D \cdot l_D} = \frac{l_{ch}}{l_D} \quad l_{ch} = \text{chip length}$$

Constant chip volume:

$$h_D \cdot b_D \cdot l_D = h_{ch} \cdot b_{ch} \cdot l_{ch}$$

Orthogonal cutting:

$$(b_D = b_{ch} = b)$$

$$r_c = \frac{h_D}{h_{ch}} = \frac{\overline{AB} \sin \phi}{\overline{AB} \cos(\phi - \gamma_0)} = \frac{\sin \phi}{\cos(\phi - \gamma_0)} \Rightarrow \tan \phi = \frac{r_c \cos \gamma_0}{1 - r_c \sin \gamma_0}$$



Velocity Diagram

Mass continuity: $v_c \cdot h_D = v_\gamma \cdot h_{ch}$

Since $h_{ch} > h_D \Rightarrow v_\gamma < v_c$.

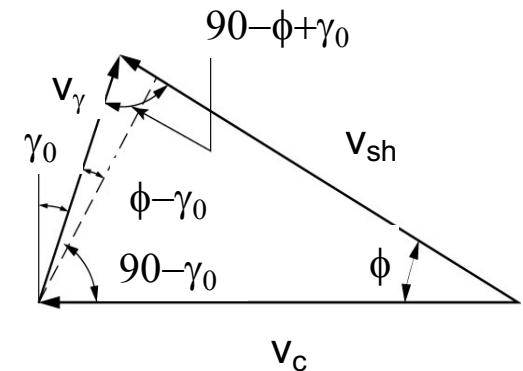
It follows that:

$$v_\gamma = v_c \frac{h_D}{h_{ch}} = v_c r_c = v_c \frac{\sin \phi}{\cos(\phi - \gamma_0)}$$

From the velocity diagram:

$$\frac{v_c}{\cos(\phi - \gamma_0)} = \frac{v_{sh}}{\cos \gamma_0} = \frac{v_\gamma}{\sin \phi}$$

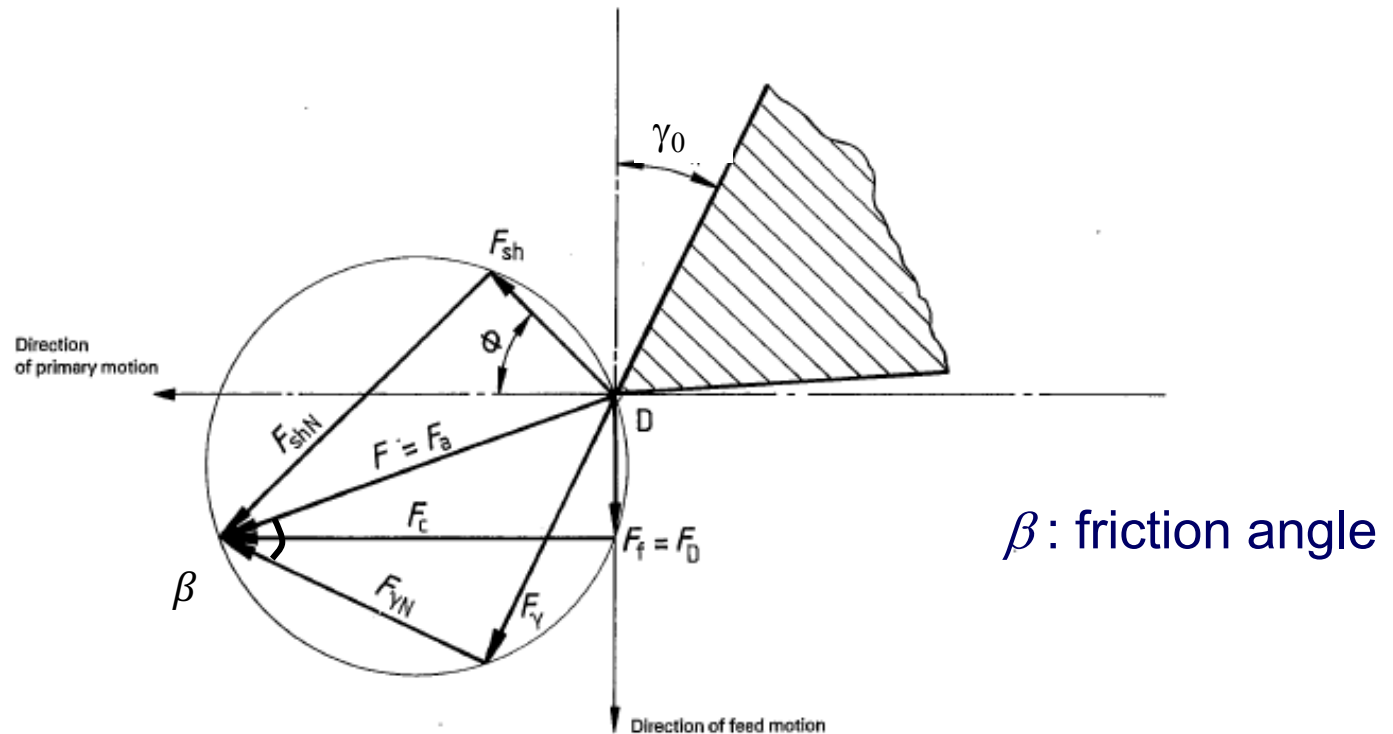
Where v_{sh} is the speed in the shear plane.





MODEL OF CHIP FORMATION

The force F represents the action of the tool on the material. It can be decomposed in different ways to fully characterize the tool-material interaction.



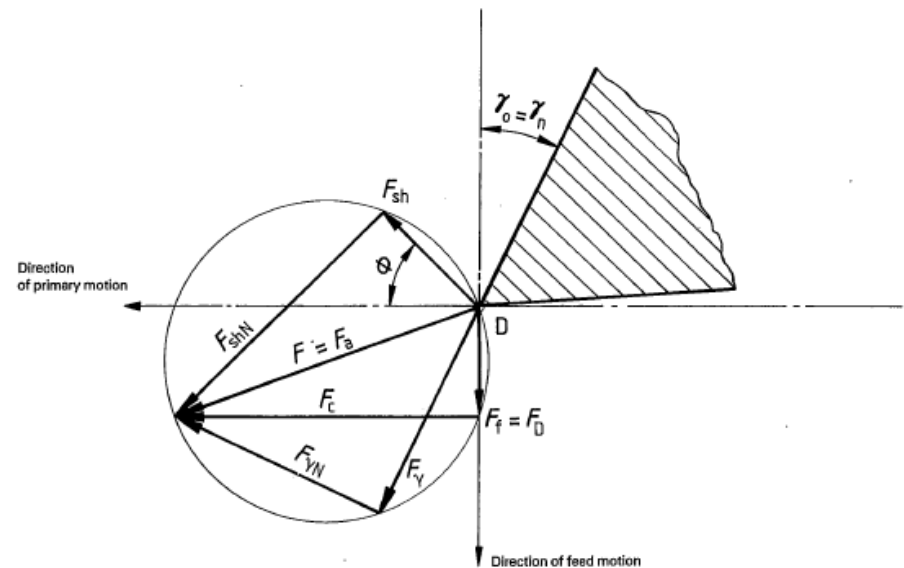
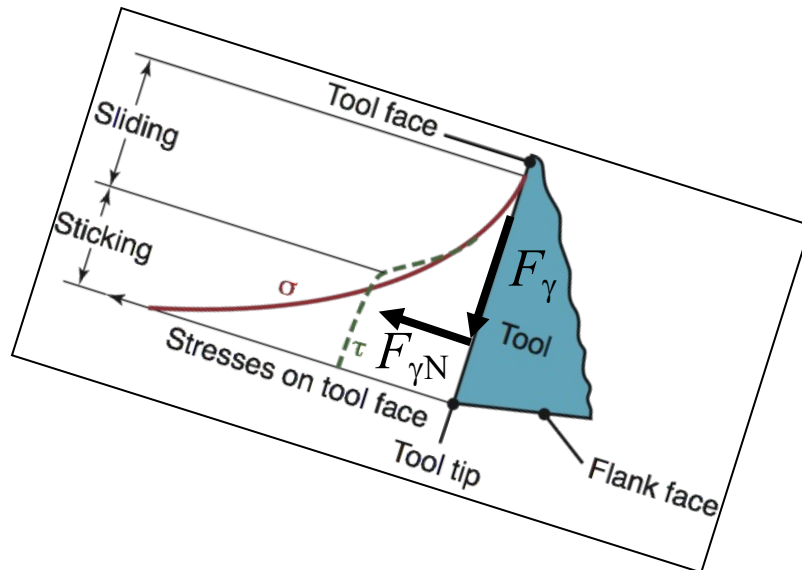


Force F_γ : parallel to the rake plane

Force $F_{\gamma N}$: normal to the rake plane

$$F_\gamma = F \sin(\beta)$$

$$F_{\gamma N} = F \cos(\beta)$$





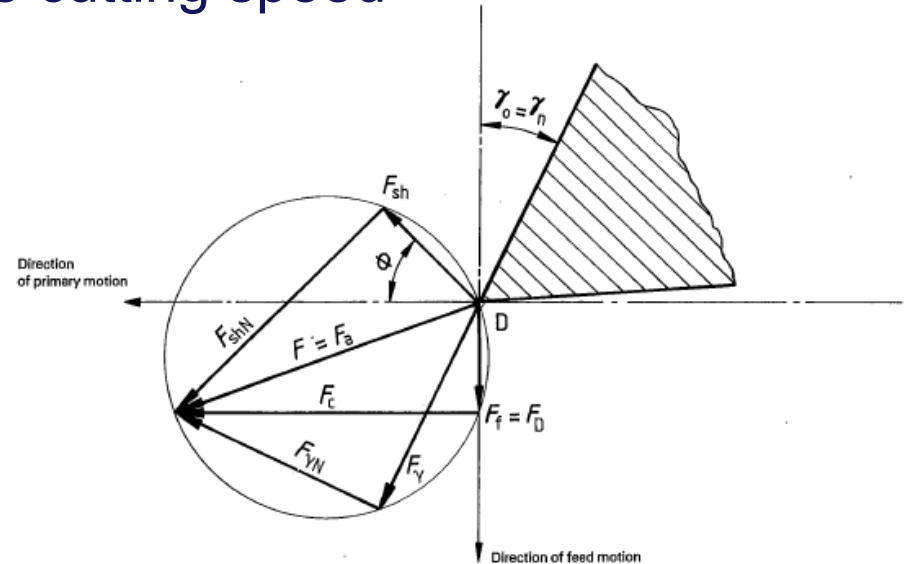
MODEL OF CHIP FORMATION

Cutting force (F_c), parallel to the cutting speed

Thrust force (F_D), normal to the cutting speed

$$F_c = F \cos(\beta - \gamma_0)$$

$$F_D = F \sin(\beta - \gamma_0)$$



These forces are not known, but they can be measured or predicted with mathematical models.



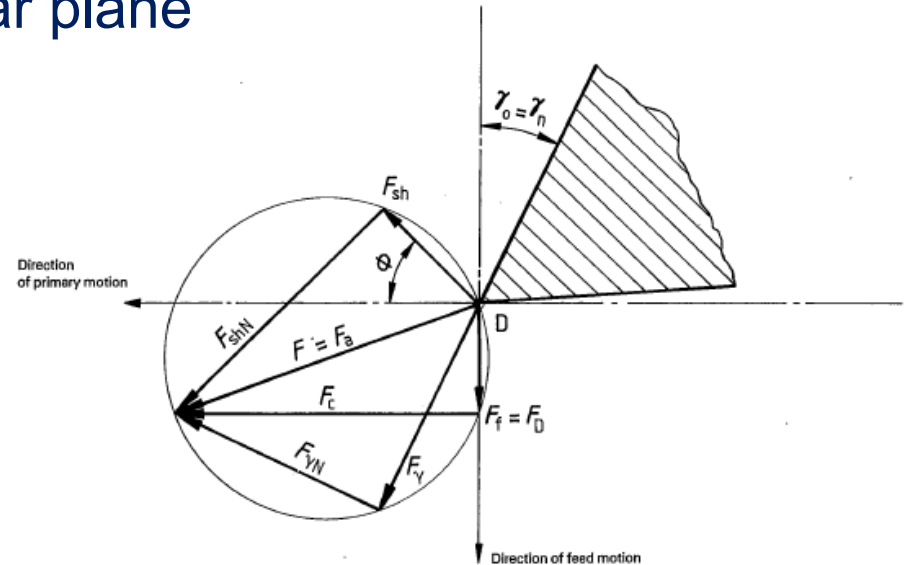
MODEL OF CHIP FORMATION

Force F_{sh} : parallel to the shear plane

Force F_{shN} : normal to the shear plane

$$F_{sh} = F \cos(\phi + \beta - \gamma_0)$$

$$F_{shN} = F \sin(\phi + \beta - \gamma_0)$$





$$\tau_{sh} = \frac{F_{sh}}{A_{sh}} = \frac{F_{sh} \sin \phi}{A_D} = \frac{F \cos(\phi + \beta - \gamma_0) \sin \phi}{A_D} = \frac{F_c \cos(\phi + \beta - \gamma_0) \sin \phi}{A_D \cos(\beta - \gamma_0)}$$

$$\sigma_{sh} = \frac{F_{shN}}{A_{sh}} = \frac{F_{shN} \sin \phi}{A_D} = \frac{F \sin(\phi + \beta - \gamma_0) \sin \phi}{A_D} = \frac{F_c \sin(\phi + \beta - \gamma_0) \sin \phi}{A_D \cos(\beta - \gamma_0)}$$

Parameters: $\gamma_0, \beta, \phi, \tau_{sh}$

- γ_0 is a property of the cutting tool
- β can be estimated:

$$\mu = \tan \beta = \frac{F_\gamma}{F_{\gamma N}} = \frac{F_D \cos \gamma_0 + F_c \sin \gamma_0}{F_c \cos \gamma_0 - F_D \sin \gamma_0} = \frac{F_D + F_c \tan \gamma_0}{F_c - F_D \tan \gamma_0}$$



$$F_c = F \cos(\beta - \gamma_0)$$

$$F_{sh} = F \cos(\phi + \beta - \gamma_0) \Rightarrow F = \frac{F_{sh}}{\cos(\phi + \beta - \gamma_0)}$$

$$F_{sh} = \tau_{sh} A_{sh} = \tau_{sh} \frac{A_D}{\sin \phi}$$



$$F_c = \frac{F_{sh} \cos(\beta - \gamma_0)}{\cos(\phi + \beta - \gamma_0)} = \tau_{sh} \frac{A_D}{\sin \phi} \frac{\cos(\beta - \gamma_0)}{\cos(\phi + \beta - \gamma_0)}$$



MODEL OF CHIP FORMATION

Pijspanen model

- No strain hardening
- No BUE (Built Up Edge)
- Plastic deformation begins when $\tau_{\max} = \tau_{\text{sh}}$ (elastic strain is not considered)
- No friction on the flank
- No friction between tool's rake and chip

Shear angle ϕ assumes a value that minimizes the shear strain.



Pijspanen model

$$\frac{\partial \delta}{\partial \phi} = \frac{\partial}{\partial \phi} [\cot \phi + \tan(\phi - \gamma_0)] = -\frac{1}{(\sin \phi)^2} + \frac{1}{(\cos(\phi - \gamma_0))^2}$$

$$\frac{(\sin \phi)^2 - (\cos(\phi - \gamma_0))^2}{[\sin \phi \cos(\phi - \gamma_0)]^2} = 0$$

$$(\sin \phi)^2 = (\cos(\phi - \gamma_0))^2 = \left[\sin \left(\frac{\pi}{2} - (\phi - \gamma_0) \right) \right]^2$$

$$\phi = \frac{\pi}{2} - (\phi - \gamma_0) \quad \Rightarrow \quad \phi = \frac{\pi}{4} + \frac{\gamma_0}{2}$$



Ernst & Merchant

- No strain hardening
- No BUE (Built Up Edge)
- No friction on the flank
- Constant τ_{sh}

Shear angle ϕ assumes a value that minimizes the cutting energy.



Ernst & Merchant

$$F_c = \tau_{sh} \frac{A_D}{\sin \phi} \frac{\cos(\beta - \gamma_0)}{\cos(\phi + \beta - \gamma_0)}$$

$$E_c = F_c L_c = \tau_{sh} \frac{A_D}{\sin \phi} \frac{\cos(\beta - \gamma_0)}{\cos(\phi + \beta - \gamma_0)} L_c$$

$$\frac{\partial E_c}{\partial \phi} = 0$$



Ernst & Merchant

$$\frac{\partial E_c}{\partial \phi} = \tau_{sh} A_D L_c \cos(\beta - \gamma_0) \frac{\partial}{\partial \phi} \left[\frac{1}{\sin \phi \cos(\phi + \beta - \gamma_0)} \right] = 0$$

$$\frac{\cos \phi \cos(\phi + \beta - \gamma_0) - \sin \phi \sin(\phi + \beta - \gamma_0)}{[\sin \phi \cos(\phi + \beta - \gamma_0)]^2} = 0$$

$$\cos(\phi + \phi + \beta - \gamma_0) = 0 = \cos \frac{\pi}{2}$$

$$2\phi + \beta - \gamma_0 = \frac{\pi}{2} \quad \Rightarrow \quad \phi = \frac{\pi}{4} + \frac{1}{2}(\gamma_0 - \beta)$$



Let's consider the Ernst & Merchant model result:

$$2\phi + \beta - \gamma_0 = \frac{\pi}{2} \rightarrow \beta - \gamma_0 = \frac{\pi}{2} - 2\phi$$

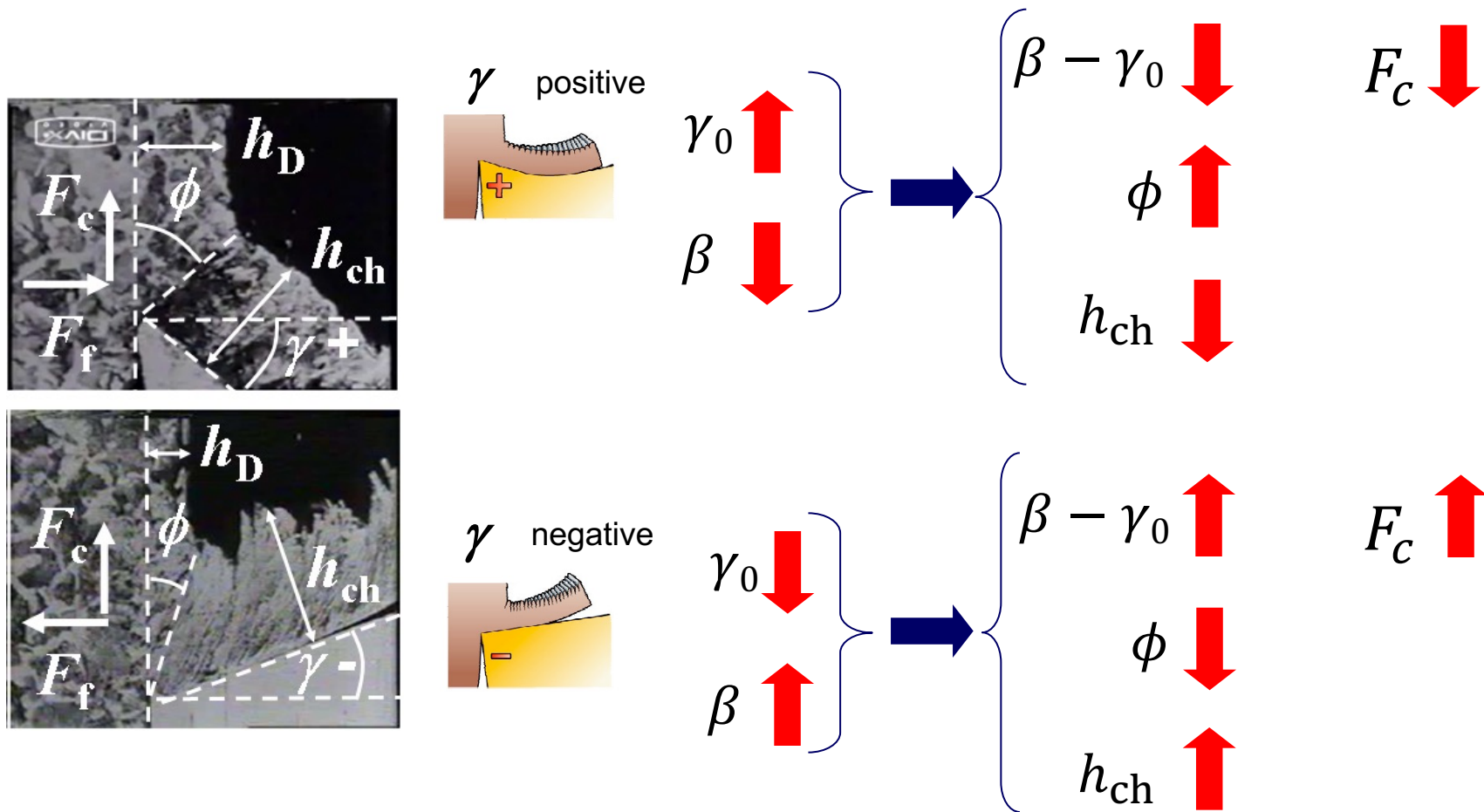
$$F_c = \frac{\tau_{sh} \cdot A_D \cdot \cos(\beta - \gamma_0)}{\sin \phi \cdot \cos(\phi + \beta - \gamma_0)} = \frac{\tau_{sh} \cdot A_D \cdot \cos(\frac{\pi}{2} - 2\phi)}{\sin \phi \cdot \cos(\phi + \frac{\pi}{2} - 2\phi)}$$

$$F_c = \frac{\tau_{sh} \cdot A_D \cdot \sin 2\phi}{\sin \phi \cdot \sin \phi} = \frac{\tau_{sh} \cdot A_D \cdot 2 \sin \phi \cos \phi}{\sin \phi \cdot \sin \phi}$$

$$F_c = \frac{2 \cdot \tau_{sh} \cdot A_D}{\tan \phi}$$



Influence of γ and β on forces

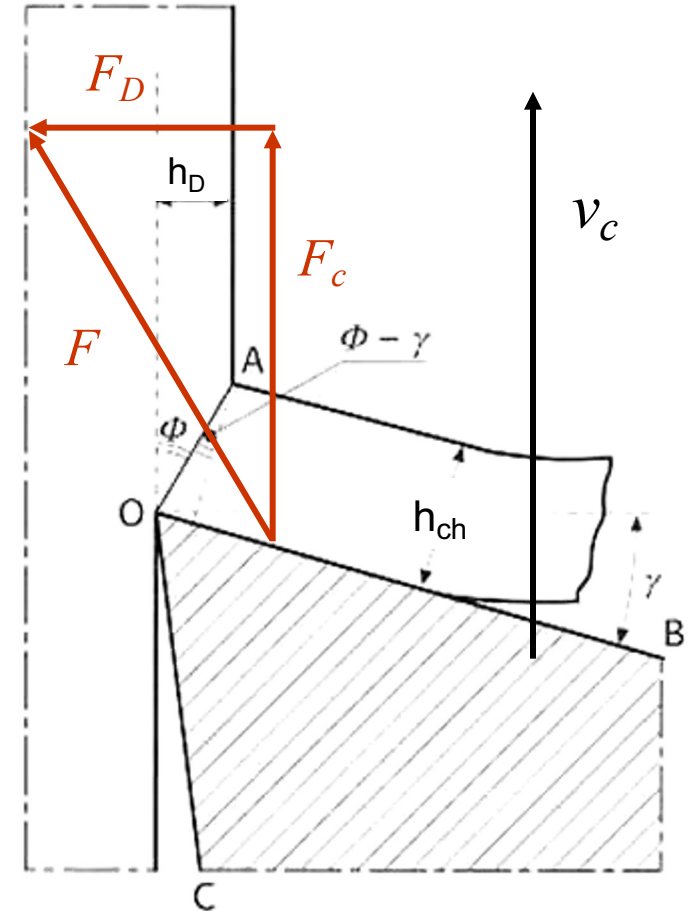




F_c is the cutting force, whose direction is parallel to the cutting speed v_c .

However the total force F in orthogonal cutting may be considered as the composition of:

- Cutting Force: F_c
- Thrust Force: F_D



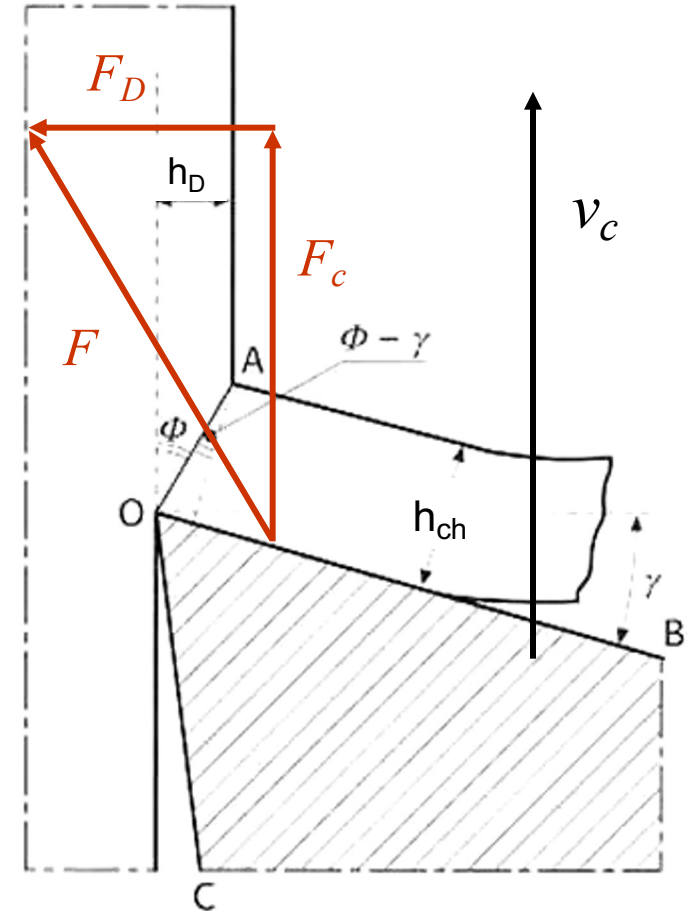


If $\vec{v}$ is the relative speed between tool and workpiece, the required power P is given by the scalar product:

$$P = \vec{F} \cdot \vec{v} = F_c v_c + F_D v_D$$

But, since $v_D = 0$, it follows that required power is the cutting power P_c :

$$P_c = F_c v_c$$





The cutting pressure k_c is defined from the relationship:

$$k_c \stackrel{\text{def}}{=} \frac{F_c}{A_D} \rightarrow F_c = k_c A_D$$

k_c depends on:

- A_D
- workpiece material
- tool material and geometry (γ_0 in particular)
- v_c
- lubrication conditions



Kronenberg relationship:

$$k_c = \frac{k_{cs}}{h_D^x \cdot b^y}$$

Where k_{cs} is the specific cutting pressure to remove a chip section of 1 mm² with $h_D = 1$ mm and $b = 1$ mm.

Typically, $y \cong 0$, then it follows that:

$$k_c = \frac{k_{cs}}{h_D^x}$$

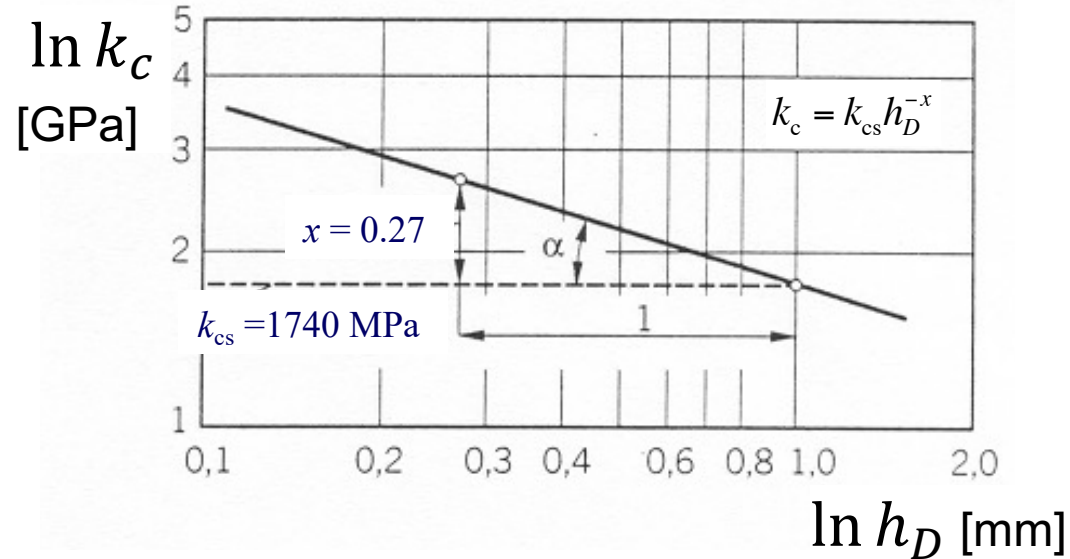
Note that:

- k_{cs} related to the material to remove;
- k_c decreases as h_D increases;
- x constant mainly related to the tool material.



$$k_c = \frac{k_{cs}}{h_D^x}$$

$$\ln k_c = -x \ln h_D + \ln k_{cs}$$



NB: k_{cs} is a value that you can find in books and catalogues



The power given by the motor is the absorbed electric power (P_e) reduced by the transformation efficiency (factor η).

Therefore, it follows that:

$$P_c \leq \eta \cdot P_e$$

In general $0,7 \leq \eta \leq 0,95$, while the maximum value P_e is a given characteristic of the machine.



CUTTING TIME AND MATERIAL REMOVAL RATE

In orthogonal cutting, the cutting time t_c can be computed as:

$$t_c = \frac{L_c}{v_c}$$

where L_c is the cutting length and v_c is the cutting speed.

It can also be computed as:

$$t_c = \frac{h_D b L_c}{h_D b v_c} = \frac{V}{MRR}$$

where:

- V is removed material volume [mm^3]
- $MRR = A_D \cdot v_c$ is the material removal rate [$\text{mm}^3 / \text{min.}$]



In orthogonal cutting, the cutting energy E_c can be computed as:

$$E_c = F_c L_c = k_c A_D L_c = k_c h_D b L_c = k_c V$$

It can also be computed as:

$$E_c = P_c t_c = F_c v_c \frac{V}{MRR} = k_c A_D v_c \frac{V}{A_D v_c} = k_c V$$

Therefore, the cutting pressure k_c can be also seen as the specific cutting energy $\left[\frac{N \cdot mm}{mm^3} \right]$ for the machining operation.